Louis Lin

Louis.Lou.Lin@gmail.com | 973-975-5359 | San Diego, California

Education

Doctor of Philosophy in Structural Engineering, *University of California, San Diego* — Expected 12/26

- Large-scale testing, finite element modeling and correlation, nonlinear dynamics of structural systems, structural vibrations, modal analysis

Master of Science in Structural Engineering, *University of California, San Diego* — 10/20 – 6/21

- Nonlinear structural analysis, advanced seismic design, based isolation and supplemental damping, advanced steel, concrete and masonry design, structural system testing and model correlation

Bachelor of Science in Civil Engineering, *Lehigh University* — 9/16 – 5/20

- Blast resistant design and analysis, structural fire engineering, building systems

Technical Experience

University of California, San Diego, *Graduate Student Researcher* — 6/21 – Present

- Led the testing of 19 specimens on four full-scale, beam-column subassemblies utilizing the DuraFuse SMF connections for new construction and retrofit following the AISC Seismic Provisions.
- Correlated two FEM of an electrical system in ABAQUS and OpenSees to run NLTHA to quantify seismic risk reduction of different designs of tuned-mass dampers or isolation devices.
- Planned, instrumented, and processed acceleration data of a full-scale shake table testing of a three-story steel frame building. Performed system identification to qualify a FEM and analytical model of the building.

Nonlinear Mechanics and Dynamics Research Institute, *Graduate Student Intern* — 7–8/24

- Investigated the phenomenon of electrical chatter and extracted modal properties of a single pin-receptacle system using laser vibrometers, tri-axial accelerometers, by performing roving hammer tests.
- Simulated simplified impact interaction through state-space formulation and nonlinear contacts in MATLAB.

California Space Grant Consortium Summer Program, *Graduate Student Lead* — 6–10/22

- Organized a technical program for 12 undergraduates using microcontrollers and 3D printing focused on aerospace related projects: miniature UAVs, solid propellant rockets, and propulsion testing stands

Alfred & Benesch, *Structural Design Intern* — 5–8/19

- Performed calculation checks on elevation and stationing for a precast reinforced-concrete bridge
- Quantified and verified placement of reinforcement bars for precast bridge footers and web pieces

Shanghai Tonghao Civil Eng. Consultation, *Bridge Design Intern* — 5–8/18

- Used Python to model and calculate road curves and elevation profile for the new roads

Pathways Development Initiatives, *Field Officer* — 5–7/17

- Created forms to evaluate loan risk for the Small Business Development program

Teaching Experience

SE 101B Mechanics II: Dynamics, *Teaching Assistant* — 1–3/24, 1–3/25, 3–6/25, 1/26–Present

- Deliver supplemental instruction to over 150 students by holding office hours, holding problem solving sessions, creating additional study material, proctoring exams, and maintaining student grades.

SE 9 Algorithms and Programming for SE, *Teaching Assistant* — 1–6/23

- Prepared and held weekly labs for 3 sections of 120 students that taught application of MATLAB to structural engineering.

SE 131B Computing Projects in SE, *Teaching Assistant* — 9–12/22

- Reviewed and provided comments on over 150 student's FEA reports for beams and shells usage in modal analysis, model reduction, composite material, buckling analysis, and topology optimizations.

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Research Experience

Prequalification of Moment Connections for SMF New Construction and Retrofit 4/25–1/26

- Orchestrated the erection of four full-scale, beam-column subassemblage that tested 19 designs for new construction and retrofitted SMF with DuraFuse moment connections following the AISC Cyclic Loading Protocol for qualifying seismic connection
- Investigated the behavior of three beam-column subassemblies in monotonic push up and pull-down tests with restored Pre-Northridge earthquake welded flange and bolted web SMF connections.

Improved Seismic Performance of Safety-Related Equipment in Nuclear Plant 12/23–12/24

- Modeled a high-fidelity FEM of an electrical component in ABAQUS and OpenSees considering linear and nonlinear hysteretic behavior of connections and global and local modal properties
- Analyzed the reduction in seismic risk for electrical cabinets through NLTHA using different designs of base isolation and tuned mass dampers devices for given floor response spectrums within a nuclear power plant.

Database of Dynamic Characteristics of Nonstructural Components 10–12/23

- Created database of dynamic properties of vibration isolated and rigidly fixed nonstructural components from AC156 shake table qualification testing reports

Shake Table Testing of Full-Scale Modular Testbed Building 10/21–10/23

- Designed and constructed six modular concrete-steel floor decks used for the building's main diaphragm. Planned and installed sensors for data acquisition spread throughout three floors.
- Conducted system identification on a full-scale three story, steel frame building with three LFRS using a series of impact tests and multi-directional white noise at UCSD's large high performance outdoor shake table.

System Identification of NCS Supported with Flexible Connection 6–10/21

- Performed modal analysis on a model nonstructural component attached with various flexible angles subjected to white noise and ground motions

Leadership Experience

Earthquake Engineering Research Institute UCSD Grad. Student Chapter, *President* 6/24–6/25

- Organized events for UCSD Masters students such as shake table visits, resume workshops, and office tours to get students acclimated to the San Diego work force and structural engineering industry
- Mentored undergraduate competition team on numerical modeling, construction, and testing of 6ft balsa wood, miniature building model subjected to a miniature shake table testing

Structural Engineering Chair's Student Council, *Member* 8/22–Present

- Presented about UCSD Structural Engineering department at external events and open houses
- Developed the Structural Engineering Survival Handbook for Graduate students

Lehigh University Seismic Design Club, *Founder & President* 8/19–5/20

- Procured materials, designed and analyzed numerical models, created the construction schedule, tested and presented the building model during the competition

Lehigh Student Chapter of Engineers Without Borders, *Executive Secretary* 8/18–5/19

- Organized communication between the executive board, advisors, and active members on ongoing water distribution project in Nicaragua

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Publications

- **Lin L**., C.M. Uang (2025) "Full-Scale Testing of DuraFuse Frames Moment Connections for SMF Applications: Series L1-L3 Specimens". TR-25/02.
- **Lin L**., C.M. Uang (2025) "Full-Scale Testing of DuraFuse Frames Moment Connections for SMF Applications: Series L4 Specimens". TR-25/01.
- **Lin L**., Kavei S., Mosqueda G., Park J. (2024) "Improved Seismic Performance of Safety Related Equipment in Nuclear Power Plants". TR-24/01.
- Mosqueda G., Kwon T.H., Erler K., Bustamante R., **Lin L.** (2023) "Improved Seismic Performance of Safety Related Equipment in Nuclear Power Plants". SSRP 23/01.
- **Lin, L**., T. Hutchinson (2024). "Fundamental Properties of Certified NCS in California Hospitals", in Fundamental Properties of Floor-Mounted Nonstructural Component in Hospitals. DesignSafe-CI. https://doi.org/10.17603/ds2-f69x-hp87
- **Lin L.**, Hutchinson T.C. (2024). "Characterizing Dynamic Properties of Floor-Mounted Nonstructural Components and Systems in Hospitals". Proceedings of the 14th World Conference on Earthquake Engineering, Milan, IT. June 30 – July 5.
- Morano M., **Lin L.**, Hutchinson T.C., Liu J., Williamson E., Pantelides C. (2024). "Modal Parameter Identification of a 3-Story BRBF Steel Building by Non-Destructive Field Testing". Earthquake Engineering and Structural Dynamics.
- Williamson E., Liu J., Morano M., **Lin L.**, Saxey B., Pantelides C., Hutchinson T.C., Mosqueda G. (2024). "Bi-directional Earthquake Response of a Three-Story Steel Structure with Buckling- Restrained Braces". Proceedings of the 14th World Conference on Earthquake Engineering, Milan, IT. June 30 – July 5.
- Morano M., Liu J., Williamson E., **Lin L**., Hutchinson T.C., Pantelides C. (2024). Measured Acceleration Amplification Derived from Shake Table Tests of A 3-Story Steel Frame Building. Proceedings of the 14th World Conference on Earthquake Engineering, Milan, IT. June 30 – July 5.
- Mosqueda G, Guerrero N., Schemmer Z., **Lin, L.,** Hutchinson T. C., Pantelides C. P. (2022). "Jupyter Notebooks for Data workflow of NHERI Experimental Facilities". Proceedings of the 12th National Conference on Earthquake Engineering, Earthquake Engineering Research Institute, Salt Lake City, UT. June 27 – July 1.
- **Lin, L**., Hutchinson, T.C., and Wang, X. (2022). System identification of a model NCS supported with flexible base connections. Proceedings of the 12th National Conference on Earthquake Engineering, Earthquake Engineering Research Institute, Salt Lake City, UT. June 27 – July 1.

Honors & Awards

DesignSafe Academy and NHERI Hackathon 1st Place	6/22
Structural Engineering Distinguished Fellowship	7/21
2021 San Diego ASCE Pipeline & Environmental Committee Scholarship	6/21
PCI Big Beam Contest 3rd Place Overall & Best Report	10/20
EERI Seismic Design Competition Spirit of the Competition	3/20
Harry Haley Ousey Scholarship	6/16–6/20

Additional

Software Skills: Python, ABAQUS, SAP2000, OpenSees, Mathcad, MATLAB, AutoCAD, Fusion 360, RISA
Work Eligibility: US Citizen
Certifications: Engineer in Training